

Project title: Youth Unemployment Despite Educational Qualifications in the IT sector. Write an academic report. Provide an abstract, literature review, project proposal, bibliography...

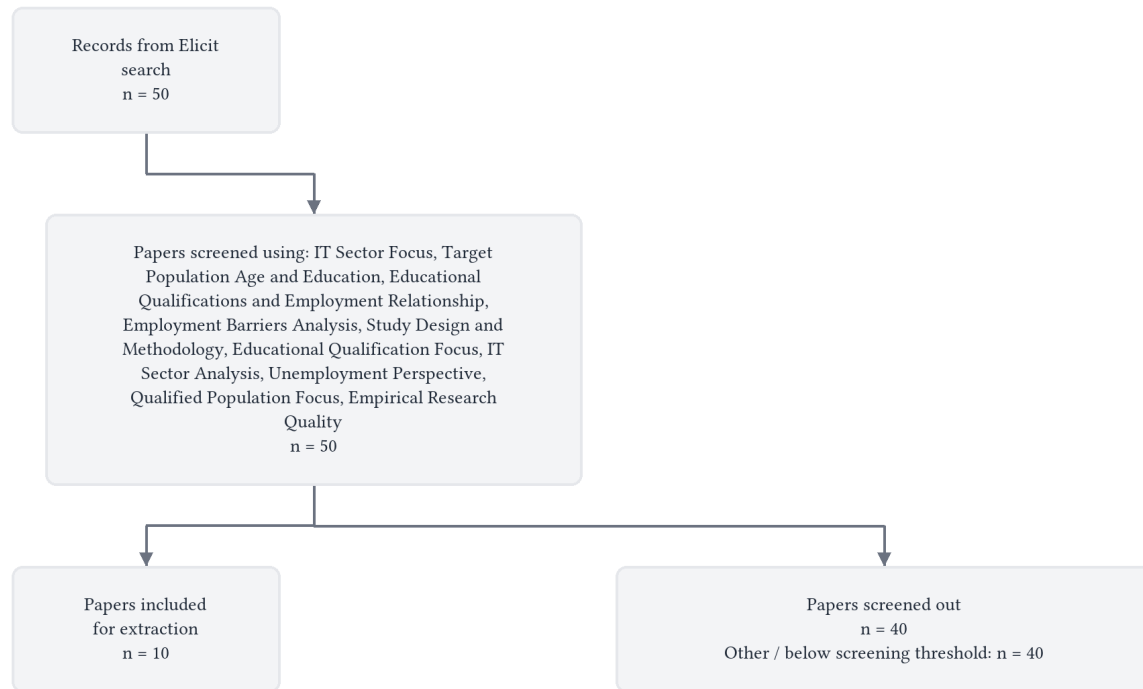
Youth unemployment despite IT qualifications results from educational systems failing to provide practical industry-aligned skills that keep pace with rapid technological change, manifesting differently across contexts as either macroeconomic job scarcity, graduate unemployability due to quality failures, or qualification-opportunity mismatches despite labor shortages.

Abstract

Youth unemployment despite educational qualifications in the IT sector manifests through distinct mechanisms across developing economies. South Africa experiences severe youth unemployment at 55.2% for ages 15-34 [1], driven primarily by macroeconomic job creation failures [2] rather than graduate inadequacy. In contrast, India demonstrates an unemployability crisis where 75-83% of IT and engineering graduates are deemed unsuitable for employment despite available positions [3, 4], reflecting fundamental educational quality failures particularly in mass-market institutions [3]. Serbia and Romania exhibit paradoxical patterns of youth unemployment rates of 32.4% [5] and 16.9% [6] respectively, while simultaneously facing acute shortages of 15,000-50,000 qualified IT professionals in Serbia [5], indicating qualification-opportunity mismatches rather than simple oversupply.

Universal contributing factors include skills gaps in technical competencies [1, 5, 7, 8], soft skills deficiencies [1, 2], and particularly critical gaps between theoretical knowledge and practical experience [1, 2, 5, 7]. Educational systems consistently fail to align curricula with employer expectations [1, 2, 7], while rapid technological change in Fourth Industrial Revolution technologies including IoT, AI, and machine learning [8] outpaces educational adaptation [1, 3, 5, 7]. Proposed solutions emphasize work-integrated learning [2, 5, 7], curriculum reforms aligned with Industry 4.0 requirements [7], and industry-academia collaboration models such as Serbia's Master 4.0 program involving partnerships with over 75 IT companies [5]. However, virtually no studies report measured outcomes from implemented interventions [1, 2, 5, 8, 9], representing a critical evidence gap preventing assessment of intervention effectiveness across contexts.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Project title: Youth Unemployment Despite Educational Qualifications in the IT sector. Write an academic report. Provide an abstract, literature review, project proposal, bibliography..."

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **IT Sector Focus:** Does the study examine unemployment, underemployment, or employment barriers specifically in the Information Technology sector?
- **Target Population Age and Education:** Does the study focus on individuals aged 15-35 years who have completed secondary, tertiary, or vocational education in IT-related fields?

- **Educational Qualifications and Employment Relationship:** Does the study examine the relationship between educational qualifications and employment outcomes in IT?
- **Employment Barriers Analysis:** Does the study investigate skill mismatches, hiring practices, or employer requirements in IT recruitment?
- **Study Design and Methodology:** Is the study a quantitative, qualitative, mixed-methods study, systematic review, or meta-analysis with empirical data (not just conference abstracts, editorials, or opinion pieces)?
- **Educational Qualification Focus:** Does the study specifically reference educational qualifications rather than focusing solely on general unemployment without qualification considerations?
- **IT Sector Analysis:** Does the study include IT sector analysis rather than being limited exclusively to non-IT sectors (healthcare, manufacturing, agriculture, etc.)?
- **Unemployment Perspective:** Does the study provide insight into unemployment or include comparison to unemployed qualified candidates (rather than examining only employed IT workers without such comparison)?
- **Qualified Population Focus:** Does the study focus on individuals with formal education or qualifications (rather than focusing exclusively on individuals without formal education or qualifications)?
- **Empirical Research Quality:** Is the study based on empirical data with methodological rigor (not limited to conference abstracts, editorials, or opinion pieces without empirical data)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 40

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Context:**

Extract contextual details about the study setting relevant to youth unemployment despite educational qualifications in the IT sector, including:

- Geographic location (country, region, city)
- Time period of study/data collection
- Specific IT sub-sectors examined (software development, cybersecurity, data analytics, etc.)
- Economic context during study period (pre/post-COVID, economic growth/recession)
- Size and maturity of local IT sector

- **Youth Demographics:**

Extract characteristics of the youth population studied in relation to unemployment despite educational qualifications in IT, including:

- Age range definition of 'youth' used in study
- Sample size of youth studied
- Educational qualifications held (degree type, institution tier, graduation year)
- Prior work experience in IT sector
- Gender, socioeconomic background if relevant

- **Unemployment Metrics:**

Extract specific unemployment and employability statistics for educated youth in the IT sector, including:

- Unemployment rate percentages for qualified youth
- Employability rate (percentage deemed employable by industry)
- Duration of unemployment after graduation
- Comparison with overall youth unemployment or other sectors
- Trends over time if available
- Definition of 'unemployed' vs 'unemployable' used

• **Skills Mismatches:**

Extract details about skills gaps that cause unemployment despite educational qualifications in IT, including:

- Technical skills lacking (programming languages, frameworks, tools)
- Soft skills deficiencies (communication, teamwork, problem-solving)
- Digital/21st century competencies missing
- Industry-specific knowledge gaps
- Practical experience vs theoretical knowledge gaps
- Assessment methods used to identify these gaps

• **Educational Quality:**

Extract information about educational provision and quality issues that contribute to graduate unemployment in IT, including:

- Type of educational institutions (elite vs mass market, public vs private)
- Curriculum relevance to industry needs
- Teaching methods and practical training availability
- Faculty qualifications and industry experience
- Infrastructure and technology resources
- Accreditation and quality assurance issues

• **Industry Requirements:**

Extract details about IT sector employment requirements and hiring challenges relevant to youth unemployment, including:

- Specific skills and competencies employers seek
- Entry-level job availability and requirements
- Employer perspectives on graduate preparedness
- Hiring practices and selection criteria
- Industry growth vs talent supply dynamics
- Regional variations in demand

• **Contributing Factors:**

Extract systemic and contextual factors that contribute to youth unemployment despite educational qualifications in IT, including:

- Economic migration and brain drain
- Information asymmetries between graduates and employers
- Rapid technological changes outpacing education
- Cultural or social barriers to employment

- Government policy impacts
- COVID-19 or other external shock effects
- Regional development disparities

- **Solutions Tested:**

Extract details about interventions, policies, or solutions implemented or proposed to address youth unemployment despite qualifications in IT, including:

- Educational reforms (curriculum updates, practical training, industry partnerships)
- Skills development programs and certification initiatives
- Government employment schemes or policies
- Industry-academia collaboration models
- Career guidance and job matching services
- Digital competency development programs
- Who implemented these solutions and their scope

- **Measured Outcomes:**

Extract quantitative and qualitative results related to youth unemployment despite educational qualifications in the IT sector, including:

- Employment rate improvements from interventions
- Skills assessment scores before/after training programs
- Employer satisfaction with graduate quality
- Long-term career progression of employed graduates
- Return on investment for educational or policy interventions
- Sustainability of employment gains
- Unintended consequences or limitations identified

Results

Characteristics of included studies

The review included 10 studies examining youth unemployment despite educational qualifications in the IT sector across multiple countries. Table 1 presents the characteristics of all included studies.

Study	Full text retrieved?	Geographic location	Study type	Sample size	Focus area
K. Ohei et al., 2019 [2]	No	South Africa [2]	Conceptual framework development [2]	Not mentioned [2]	ICT graduate employability framework
D. Singh & Geetali Tilak, 2020 [7]	Yes	Pune City, India [7]	Mixed methods survey [7]	50 engineering students, 20-25 years [7]	Education 4.0 model implementation

Study	Full text retrieved?	Geographic location	Study type	Sample size	Focus area
Abiodun Alao et al., 2024 [8]	Yes	South Africa (Cape Town, Johannesburg, Pretoria, Limpopo) [8]	Quantitative survey [8]	140 respondents aged 18+ [8]; 55% female, 45% male [8]	ICT skills and gender disparity in youth unemployment
R. Misra & Khushbu Khurana, 2017 [10]	No	Not specified [10]	Literature review [10]	Not applicable [10]	Employability skills for IT professionals
K. Ohei & Evangelos Mantzaris, 2023 [1]	Yes	South Africa [1]	Critical literature review [1]	Youth aged 15-34 [1]	ICT skills as enablers for sustainable youth employability
P. Choudhary & V. Chadha, 2017 [3]	No	India [3]	Conceptual review [3]	Not mentioned [3]	Unemployment and unemployability among engineers
P. Vanitha & A. Jaganathan, 2016 [4]	No	India [4]	Literature review [4]	Over 150,000 engineering students graduated in 2015 [4]	Factors affecting engineering graduate unemployment
C. Barna & M. Epure, 2020 [6]	No	Romania [6]	Policy analysis [6]	Not mentioned [6]	Youth unemployment and digital literacy during COVID-19
Nidhi Sehgal & S. Nasim, 2017 [9]	No	India [9]	Conceptual review [9]	Not mentioned [9]	Predictors of graduate employability in IT sector
Valentina Vukmirović et al., 2021 [5]	Yes	Serbia [5]	Literature review and policy analysis [5]	Youth aged 15-24 [5]	21st century digital skills development

The studies spanned multiple developing economies including South Africa [1, 2, 8], India [3, 4, 7, 9], Romania [6], and Serbia [5], examining the phenomenon during different time periods from 2016 to 2024. Data collection occurred around 2023 for the most recent South African study [8], during 2020 for the Romanian study coinciding with COVID-19 [6], and during the COVID-19 pandemic for the Serbian study which noted IT sector expansion in 2020 [5]. Most studies were conceptual or literature reviews rather than primary empirical investigations, though three studies (D.

Singh & Geetali Tilak, 2020; Abiodun Alao et al., 2024; Valentina Vukmirović et al., 2021) conducted primary data collection. Full texts were available for only 4 of the 10 studies, while 6 were analyzed based on abstracts alone, which may limit the depth of extracted information.

Unemployment magnitude and employability statistics

Table 2 presents unemployment and employability statistics from studies that reported quantitative metrics.

Study	Geographic context	Youth unemployment rate	Employability assessment	Age group	Comparison metrics
K. Ohei & Evangelos Mantzaris, 2023 [1]	South Africa	55.2% in Q1 2019 [1]; peaked at 56.40% in Q2 2019 [1]	Not measured [1]	15-34 years [1]	Youth account for 63.4% of total unemployed population [1]; increased from 27.2% in 2018 to 27.5% in 2019 [1]
P. Choudhary & V. Chadha, 2017 [3]	India	Not measured [3]	83% of engineering graduates unemployable [3]; 75% of IT graduates unemployable [3]	Not specified [3]	National Employability Report 2012 [3]; NASSCOM study [3]
P. Vanitha & A. Jaganathan, 2016 [4]	India	Not measured [4]	80% of engineering graduates unemployable [4]	Graduates from 2015 [4]	Based on Aspiring Minds National Employability Report [4]
C. Barna & M. Epure, 2020 [6]	Romania	16.9% youth unemployment vs. 5.3% national unemployment in June 2020 [6]	Not measured [6]	Young people (age not specified) [6]	Youth unemployment rate 3.2 times higher than national rate [6]
Valentina Vukmirović et al., 2021 [5]	Serbia	32.4% [5]	Not measured [5]	15-24 years [5]	IT sector expanded despite overall economic downturn during COVID-19 [5]

The data reveal striking disparities in youth unemployment across contexts, with South Africa experiencing the

highest rate at 55.2% for the 15-34 age group in early 2019 [1]. Youth unemployment in South Africa peaked at 56.40% in Q2 2019 [1] and showed an upward trend from 27.2% in 2018 to 27.5% in 2019 [1]. Serbia reported 32.4% youth unemployment for ages 15-24 [5], while Romania's youth unemployment rate of 16.9% in June 2020 was 3.2 times higher than the national unemployment rate of 5.3% [6].

The Indian context presented a different metric, focusing on unemployability rather than unemployment. Multiple studies reported that 75-83% of engineering and IT graduates were deemed unemployable by industry standards [3, 4]. Specifically, the National Employability Report by Aspiring Minds in 2012 found 83% of engineering graduates unfit for employment [3], while a NASSCOM study indicated 75% of IT graduates were unemployable [3]. A 2016 study based on data from over 150,000 engineering students who graduated in 2015 from over 650 colleges found 80% were unemployable [4, 4]. These figures suggest that the problem in India is not primarily about finding employment opportunities, but rather about graduates lacking the qualifications that make them suitable for available positions.

Thematic analysis

Skills mismatches between education and industry requirements

All studies identified significant gaps between the skills graduates possess and those required by IT sector employers. Table 3 summarizes the skills deficiencies across studies.

Study	Technical skills gaps	Soft skills deficiencies	Digital/21st century competencies	Practical vs. theoretical gaps
K. Ohei et al., 2019 [2]	Not specified [2]	Lack of soft skills [2]	Not mentioned [2]	Lack of practical experience [2]
D. Singh & Geetali Tilak, 2020 [7]	IT companies find graduates lacking necessary technical skills [7]	Need for more practical and project-based learning [7]	Education needs to align with Industry 4.0 standards [7]	Lack of exposure to industries and live projects [7]
Abiodun Alao et al., 2024 [8]	ICT technical skills, programming languages, frameworks, tools [8]	Not mentioned [8]	Digital literacy [8]	Gap between theoretical knowledge and practical experience [8]
K. Ohei & Evangelos Mantzaris, 2023 [1]	Proficiency in Microsoft Office, programming languages, frameworks, tools [1]	Problem-solving, creativity, interpersonal communication [1]	Ability to access and use ICTs effectively [1]	Emphasis on hands-on skills training [1]
Valentina Vukmirović et al., 2021 [5]	Advanced digital skills such as software use and problem-solving [5]	Not explicitly mentioned [5]	Lack of digital competencies among Serbian youth [5]	Lack of staff trained in digital environments [5]

Technical skills deficiencies were consistently identified across contexts. Studies highlighted gaps in programming languages, frameworks, and tools [1, 8], proficiency in Microsoft Office [1], and advanced digital skills including

software use and problem-solving [5]. In India, IT companies reported that graduates lacked necessary technical skills despite formal qualifications [7]. The South African studies emphasized the need for ICT technical skills aligned with Fourth Industrial Revolution (4IR) technologies including IoT, AI, and machine learning [8].

Soft skills deficiencies represented another critical gap. Studies identified lacks in problem-solving, creativity, and interpersonal communication [1]. One study emphasized the need for more practical and project-based learning approaches to develop these competencies [7]. The absence of soft skills was linked to both business skills and hard skills deficiencies [2].

Digital and 21st century competencies emerged as a third major gap area. Serbian youth demonstrated a lack of digital competencies [5], while South African studies highlighted deficiencies in digital literacy [8] and the ability to access and use ICTs effectively [1]. The Serbian context revealed employers seeking specific competencies including programmatic trading, analytical skills, and cross-media skills [5], suggesting rapid evolution in required digital competencies.

The practical experience versus theoretical knowledge gap was identified as particularly problematic. Graduates lacked exposure to industries and live projects [7], hands-on skills training [1], and experience in digital environments [5]. Studies emphasized that the gap between theoretical knowledge and practical experience in ICT skills significantly impacted employability [2, 8].

Industry-specific knowledge gaps were noted in several studies. South African research identified shortages in highly sought-after ICT professions including Mobile Application Developer and Web Designer [1]. Employers in Serbia sought skills in emerging technologies such as AI, blockchain, and mobile advertising [5]. The rapid pace of technological change meant that educational systems struggled to keep curricula aligned with evolving industry needs [7].

Educational quality and curriculum relevance issues

Studies consistently identified educational provision as a key contributor to graduate unemployability. Table 4 presents educational quality issues across studies.

Study	Curriculum relevance issues	Teaching methods problems	Practical training gaps	Industry alignment
K. Ohei et al., 2019 [2]	Curriculum should be restructured to align with employer requirements [2]	Lack of soft and hard skills development [2]	Work-integrated learning recommended [2]	Need for proper communication among stakeholders [2]
D. Singh & Geetali Tilak, 2020 [7]	Curriculum lacks practical relevance and industry-based assignments [7]	Limited practical training and industry exposure [7]	Not mentioned [7]	Poor faculty development programs and selection processes [7]
Abiodun Alao et al., 2024 [8]	Educational systems should provide skills demanded by job market [8]	Targeted skill development interventions needed [8]	ICT skills education emphasized [8]	Collaboration between government and private organizations recommended [8]

Study	Curriculum relevance issues	Teaching methods problems	Practical training gaps	Industry alignment
K. Ohei & Evangelos Mantzaris, 2023 [1]	Universities not aligning curriculum with employer expectations [1]	Limited educational value and lack of skills advancement [1]	Not mentioned [1]	Recommendations for using ICTs to improve skill advancement [1]
P. Choudhary & V. Chadha, 2017 [3]	High percentage of unemployable graduates suggests lack of curriculum relevance [3]	Not explicitly mentioned but implied inadequate [3]	Not mentioned [3]	Particularly problematic in private sector institutions [3]
Valentina Vukmirović et al., 2021 [5]	Gap in curriculum relevance to industry needs [5]	Practical training available through partnerships with IT companies [5]	Not mentioned [5]	Master 4.0 program aims to bridge gap [5]

Curriculum misalignment with industry needs emerged as a fundamental problem. Universities failed to align their curricula with employer expectations [1], resulting in graduates lacking skills demanded by the job market [8]. The curriculum lacked practical relevance and industry-based assignments [7], and needed restructuring to align with employer requirements [2]. This gap was particularly pronounced in the context of Industry 4.0, where education systems remained aligned with Industry 2.0 standards [7].

Teaching methods were identified as inadequate for developing necessary competencies. Studies noted limited educational value and lack of skills advancement [1], limited practical training and industry exposure [7], and the need for targeted skill development interventions [8]. Faculty quality issues compounded these problems, with poor faculty development programs and selection processes documented in the Indian context [7].

The absence of practical training was highlighted repeatedly. Work-integrated learning (WIL) was recommended as a best practice [2], and ICT skills education was emphasized as necessary for employability [8]. The Serbian Master 4.0 program demonstrated one approach to addressing this gap through partnerships with over 75 IT companies for tutoring and internship programs [5].

Institutional quality variations contributed to disparities in graduate employability. In India, the majority of technical institutes, particularly in the private sector, produced unemployable graduates, while a few elite institutions were exceptions [3]. This suggests that the problem of youth unemployment despite qualifications is not uniform across all educational institutions, but concentrated in mass-market providers.

Industry requirements and hiring challenges

Employers' perspectives on graduate preparedness and their hiring requirements revealed substantial gaps. Table 5 summarizes industry requirements across studies.

Study	Specific competencies sought	Hiring challenges	Graduate preparedness assessment
K. Ohei et al., 2019 [2]	Soft and hard skills, business skills [2]	Shortage of jobs due to socio-economic problems [2]	Graduates lack necessary skills [2]
D. Singh & Geetali Tilak, 2020 [7]	Skills relevant to Industry 4.0, practical knowledge [7]	Significant skills gap posing hiring challenges [7]	Graduates lack practical knowledge and skills [7]
Abiodun Alao et al., 2024 [8]	ICT technical skills, expertise in IoT, robotics, AI, ML [8]	Educational systems need to equip graduates with market-demanded skills [8]	Not mentioned [8]
R. Misra & Khushbu Khurana, 2017 [10]	Technical skills, higher order thinking, personal skills, social skills [10]	Gaps between academic skills and employer requirements [10]	Not mentioned [10]
K. Ohei & Evangelos Mantzaris, 2023 [1]	Proficiency in Microsoft Office, programming languages, web design [1]	Shortage of highly skilled professionals globally [1]	Employers skeptical about preparedness due to cultural and communication barriers [1]
P. Choudhary & V. Chadha, 2017 [3]	Not specified [3]	Employers face challenges finding skilled personnel [3]	Graduates considered unemployable [3]
Nidhi Sehgal & S. Nasim, 2017 [9]	Technical skills, organisational knowledge, personal and interpersonal skills [9]	High manpower requirements but challenge finding employable workforce [9]	Not mentioned [9]
Valentina Vukmirović et al., 2021 [5]	Information and analytical skills, cross-media competencies, AI, blockchain skills [5]	Employers struggle to find suitable candidates [5]	Lack of digital competences among graduates [5]

Employers consistently sought a combination of technical and soft skills. Technical requirements included proficiency in Microsoft Office [1], programming languages [1], web design [1], and expertise in emerging technologies such as IoT, robotics, AI, and machine learning [8]. In Serbia, employers specifically sought information and analytical skills, cross-media and social media competencies, and skills in technologies such as AI, blockchain, and mobile advertising [5]. These requirements reflect the rapid evolution of the IT sector and the Fourth Industrial Revolution context [7, 8].

Soft skills requirements were equally emphasized. Studies identified the need for higher order thinking skills, personal skills, and social skills [10], as well as organisational knowledge and interpersonal skills [9]. Problem-solving, creativity, and interpersonal communication were specifically highlighted as essential for job market success [1].

Hiring challenges stemmed from multiple factors. Employers preferred hiring graduates with existing digital skills rather than providing in-house training [1], creating a barrier for graduates without practical experience. The global shortage of highly skilled professionals [1] meant that competition for qualified candidates was intense. In India,

despite high manpower requirements in the IT sector, employers faced significant challenges finding an employable workforce [9]. Serbian employers reported that over 40% faced difficulties recruiting ICT specialists [5].

Employers' assessment of graduate preparedness was largely negative. Graduates were perceived as lacking necessary skills [2, 7], and employers were skeptical about their preparedness due to cultural and communication barriers [1]. The perception that 75-83% of IT and engineering graduates were unemployable in India [3, 3] reflected fundamental misalignment between educational outputs and industry requirements.

The talent supply dynamics revealed paradoxes across contexts. While the IT sector in Serbia expanded during COVID-19 and achieved an export surplus in 2020 [5, 5], it simultaneously recorded a deficit in available professionals [5]. Serbia faced an estimated shortage of 15,000 to 50,000 professionals in the short term [5]. This suggests that unemployment among qualified youth coexisted with labor shortages, indicating a mismatch problem rather than a simple oversupply of graduates.

Systemic and contextual contributing factors

Beyond skills mismatches and educational quality issues, studies identified broader systemic factors contributing to youth unemployment despite qualifications. Table 6 presents these factors.

Study	Economic factors	Technological disruption	Information asymmetries	Policy impacts	Regional disparities
K. Ohei et al., 2019 [2]	Socio-economic problems leading to job shortages [2]	Not mentioned [2]	Information asymmetries between graduates and employers [2]	Not mentioned [2]	Not mentioned [2]
D. Singh & Geetali Tilak, 2020 [7]	Lack of skills among youngsters [7]	Rapid technological changes outpacing education [7]	Mismatch between educational accomplishments and required skills [7]	Not mentioned [7]	Not mentioned [7]
Abiodun Alao et al., 2024 [8]	Economic gender disparities and limited opportunities [8]	Rapid technological changes outpacing education [8]	Limited access to resources and insufficient ICT skills education [8]	Government policy impacts on ICT skills gaps [8]	Disparities in social class leading to unequal digital literacy [8]
K. Ohei & Evangelos Mantzaris, 2023 [1]	Not mentioned [1]	Impact of 4IR on job scarcity [1]	Universities not aligning curriculum with employer expectations [1]	Government promoting ICT skills [1]	Limited access to technology and socio-economic barriers [1]

Study	Economic factors	Technological disruption	Information asymmetries	Policy impacts	Regional disparities
P. Choudhary & V. Chadha, 2017 [3]	Reckless expansion of technical education institutions [3]	Rapid technological changes outpacing education [3]	Information asymmetries between graduates and employers [3]	Government policy impacts [3]	Not mentioned [3]
C. Barna & M. Epure, 2020 [6]	Not mentioned [6]	Rapid technological changes outpacing education [6]	Not mentioned [6]	Government policy impacts [6]	Regional development disparities [6]
Valentina Vukmirović et al., 2021 [5]	Economic migration and brain drain [5]	Rapid technological changes [5]	Employers struggle to find suitable candidates; young people lack information about labor market needs [5]	Need to harmonize educational policies [5]	Focus on Serbia; need for foreign investment [5]

Economic and socio-economic factors played significant roles. In South Africa, socio-economic problems resulted in shortage and non-creation of jobs to accommodate up-and-coming ICT graduates [2]. Gender disparities created economic barriers, with economic gender disparities and limited opportunities due to traditional gender roles and discrimination noted in South Africa [8]. Economic migration and brain drain reduced the supply of qualified professionals in Serbia, further reducing the availability of skilled workers in the domestic labor market [5].

Rapid technological change consistently outpaced educational adaptation. The Fourth Industrial Revolution (4IR) exacerbated job scarcity and inequality due to scarcity of advanced ICT skills [1]. Occupations evolved quickly, and educational achievements did not meet required skills [7]. The rapid evolution of work environments due to technological advancements and automation contributed to youth unemployment [8]. This technological disruption created a moving target for educational institutions, which struggled to keep pace with industry evolution [3, 6, 7].

Information asymmetries between graduates and employers represented a critical barrier. Universities failed to align their curricula with employer expectations [1], and there was a mismatch between educational accomplishments and skills required in industry [7]. Young people felt insufficiently informed about labor market needs [5], while employers struggled to find suitable candidates despite high unemployment rates [5]. These information gaps prevented efficient matching of qualified graduates with available positions [2, 3, 8].

Government policy impacts were noted in several contexts. The reckless expansion of technical education institutions in India, driven by policy choices prioritizing quantity over quality, contributed to unemployability [3]. Government promotion of ICT skills and addressing digital skills gaps represented attempts to mitigate the problem [1]. The need to harmonize educational policies to attract foreign investment was identified in Serbia [5]. These policy dimensions highlighted the role of government decisions in either exacerbating or ameliorating graduate unemployment.

Regional development disparities created unequal access to opportunities. Disparities in social class and environment led to unequal digital literacy in South Africa [8]. Limited access to technology and socio-economic barriers affected different populations unequally [1]. Romania's low ranking (27th out of 28 EU Member States) in the Digital Economy and Society Index in 2019 [6] reflected systemic regional disadvantages in digital competitiveness.

COVID-19 pandemic effects were documented in two studies. Romania's study occurred in June 2020 during accelerated digital transformation caused by the COVID-19 crisis [6]. Serbia experienced a reduction in labor market supply and demand during COVID-19 in numerous economic areas, though the IT sector expanded and achieved an export surplus in 2020 [5]. The pandemic both disrupted labor markets and accelerated digital transformation, creating complex effects on youth employment in IT sectors.

Proposed and implemented interventions

Studies proposed various interventions to address youth unemployment despite qualifications, though few reported implementation outcomes. Table 7 summarizes proposed solutions.

Study	Educational reforms	Skills development programs	Industry-academia collaboration	Digital competency initiatives
K. Ohei et al., 2019 [2]	Curriculum restructuring to align with employer requirements [2]	Not mentioned [2]	Work-integrated learning (WIL) [2]	Not mentioned [2]
D. Singh & Geetali Tilak, 2020 [7]	Education 4.0 model aligned with Industry 4.0 [7]	Formative evaluation including problem-solving testing [7]	Industry-based assignments and work-coordinated learning [7]	Implied through Industry 4.0 focus [7]
Abiodun Alao et al., 2024 [8]	ICT technical education [8]	ICT skills development initiatives [8]	Inclusive workplaces embracing technological advancements [8]	Provision of ICT resources to educational institutions [8]
K. Ohei & Evangelos Mantzaris, 2023 [1]	Government promoting digital literacy [1]	ICT skills development within DHET and research institutions [1]	Community programs and education institutions [1]	Promoting digitization in grassroots areas [1]
Valentina Vukmirović et al., 2021 [5]	Master 4.0 program involving partnerships with 75+ IT companies [5]	Master 4.0 program developing practical knowledge [5]	Partnerships with IT companies for tutoring and internships [5]	Focus on developing digital competencies through formal education [5]

Educational reforms centered on curriculum alignment with industry needs. The Education 4.0 model proposed for India emphasized personalized education aligned with Industry 4.0 requirements [7], including industry-based assignments and work-coordinated learning [7]. In South Africa, curriculum restructuring to align with employer requirements was recommended [2], along with government promotion of digital literacy among youth [1]. These reforms aimed to bridge the gap between theoretical education and practical industry requirements.

Work-integrated learning emerged as a key strategy. WIL was recommended as a best practice for improving ICT graduate employability in South Africa [2]. Industry-based assignments and work-coordinated learning were proposed as part of the Education 4.0 framework in India [7]. Serbia's Initiative digital Serbia implemented the Master 4.0 educational program, which involved partnerships with over 75 IT companies for tutoring and internship programs [5], providing a concrete example of industry-academia collaboration in practice.

Skills development programs focused on ICT competencies. In South Africa, policymakers were urged to implement ICT skills development initiatives to mitigate youth unemployment [8], with recommendations to utilize ICTs within the Department of Higher Education and Training (DHET) and research institutions to improve skill advancement [1]. The Master 4.0 program in Serbia aimed to develop practical knowledge needed in work environments [5].

Digital competency development programs targeted foundational digital skills. South African initiatives focused on provision of ICT resources to educational institutions for skills training [8] and promoting digitization in grassroots areas through government-sponsored initiatives [1]. Serbia emphasized developing digital competencies through formal education [5].

Industry-academia collaboration models varied in specificity and scope. Recommendations ranged from general calls for inclusive workplaces embracing technological advancements [8] to specific programs like Serbia's Master 4.0 partnerships with IT companies for tutoring and internships [5]. Community programs and education institutions were recommended in South Africa to enhance access to technology [1].

Implementation details and measured outcomes were largely absent from the studies. Most proposals remained at the conceptual or recommendation stage, with little evidence of systematic evaluation. The Education 4.0 framework proposed for India reported that 88% of respondents believed implementation would improve graduate quality [7], but this represented expectations rather than measured outcomes. No studies reported employment rate improvements, skills assessment scores before and after interventions, employer satisfaction changes, long-term career progression data, return on investment calculations, or sustainability of employment gains [1, 2, 4–6, 8–10].

Synthesis

The findings reveal a consistent pattern of youth unemployment despite IT qualifications across diverse developing economy contexts, but the underlying mechanisms and severity differ substantially by regional and economic context. This heterogeneity can be explained through four key dimensions.

Context-specific unemployment dynamics: South Africa exhibited the highest youth unemployment rates at 55.2% for ages 15-34 [1], fundamentally driven by macroeconomic job creation failures [2]. Youth accounted for 63.4% of the total unemployed population [1], indicating structural economic problems beyond skills mismatches. In contrast, India demonstrated an unemployability crisis rather than an unemployment crisis, with 75-83% of IT and engineering graduates deemed unsuitable for employment despite available positions [3, 4]. Serbia and Romania showed moderate youth unemployment rates of 32.4% [5] and 16.9% [6] respectively, but faced acute shortages of 15,000-50,000 qualified IT professionals in Serbia [5], revealing a paradoxical coexistence of youth unemployment and labor shortages.

This apparent contradiction—high youth unemployment alongside labor shortages—resolves when examining the employability versus unemployment distinction. In South Africa, socio-economic problems resulted in non-creation of jobs [2], meaning qualified graduates could not find positions even if fully prepared. In India, the issue was unemployability—graduates could not secure positions because employers found them lacking necessary competencies [3]. In Serbia, the IT sector expanded during COVID-19 [5] but could not fill positions due to insufficient qualified candidates [5]. These represent fundamentally different problems requiring distinct solutions.

Differential skills gap patterns: Technical skills deficiencies were universal, but their specific manifestations varied by technological context and educational system maturity. In India, where the IT sector is well-established, employers reported graduates lacked basic necessary technical skills despite formal credentials [7], suggesting fundamental educational quality failures. In South Africa, the gap centered on advanced ICT skills required for Fourth Industrial Revolution technologies including IoT, AI, and machine learning [1, 8], indicating a challenge of keeping pace with technological frontier advancement. Serbia identified needs for emerging technology skills in AI, blockchain, and mobile advertising [5], representing cutting-edge competency requirements.

The practical versus theoretical knowledge gap exhibited remarkable consistency across all contexts studied. Graduates universally lacked exposure to industries and live projects [7], hands-on skills training [1], and experience in digital environments [5]. This suggests that while specific technical competencies vary by regional IT sector development, the fundamental problem of experiential learning deficits transcends geographic boundaries. The persistence of this gap despite its recognition across contexts indicates systemic barriers to implementing work-integrated learning at scale.

Educational system quality stratification: The finding that India's elite institutions produced employable graduates while mass-market institutions (particularly private sector) produced 75-83% unemployable graduates [3, 3] reveals a quality stratification problem. The reckless expansion of technical education institutions in India prioritized quantity over quality [3], creating a two-tier system. This stratification helps explain why high unemployability rates coexist with a thriving IT services sector—elite institution graduates fill available positions while mass-market graduates remain unemployable.

Romania's ranking of 27th out of 28 EU Member States in the Digital Economy and Society Index [6] suggests systemic national-level educational disadvantages compared to other European countries. South Africa's failure to align university curricula with employer expectations [1] indicates institutional misalignment rather than resource constraints alone. These findings suggest that educational quality problems operate at multiple levels: individual institutional quality, national educational system effectiveness, and broader regional competitiveness.

Intervention effectiveness and implementation gaps: Despite extensive proposals for educational reforms, work-integrated learning, and industry-academia collaboration across studies [1, 2, 5, 7, 8], measured outcomes from implemented interventions were virtually absent [1, 2, 4–6, 8–10]. Serbia's Master 4.0 program represented the most concrete implementation example, involving partnerships with over 75 IT companies for tutoring and internships [5], but even this lacked reported employment outcomes or effectiveness metrics.

The gap between proposed solutions and measured implementation reveals a critical limitation in the literature. Studies identifying problems and proposing solutions outnumbered studies evaluating solution effectiveness by a large margin. The high percentage (88%) of Indian respondents believing Education 4.0 implementation would improve outcomes [7] represents expectations rather than evidence. This implementation and evaluation gap means the field lacks empirical guidance on which interventions work, in which contexts, and why.

Technological disruption as a moving target: The consistent finding that rapid technological changes outpaced educational adaptation [1, 3, 5–8] suggests a fundamental challenge beyond curriculum reform alone. The Fourth Industrial Revolution created demands for skills in IoT, robotics, AI, and machine learning [8] that educational institutions struggled to incorporate. Serbia's identification of employer needs for AI, blockchain, and mobile advertising skills [5] represented cutting-edge requirements that would have been irrelevant five years earlier.

This creates a structural problem: by the time educational institutions update curricula and train faculty in new technologies, industry requirements may have evolved further. The South African emphasis on employers preferring to hire graduates with existing digital skills rather than providing in-house training [1] exacerbates this problem,

placing full responsibility for technological currency on educational institutions that inherently lag industry in technology adoption.

Synthesis conclusion: Youth unemployment despite IT qualifications results from distinct mechanisms across contexts—macroeconomic job creation failure in South Africa, educational quality collapse in India's mass-market institutions, and qualification-opportunity mismatch in Serbia and Romania. Skills gaps are universal but manifest differently depending on local IT sector maturity and technological frontier position. Educational system stratification creates pockets of success within broader failure patterns. The near-total absence of rigorous intervention evaluations represents a critical knowledge gap preventing evidence-based policy formation. Rapid technological change creates a structural challenge where educational institutions perpetually lag industry requirements, suggesting that solutions must build adaptive capacity rather than merely updating static curricula to current needs.

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